

MB Note

1 Dinamical Systems

1.1 Models of Biological Systems

Logistic Growth Model (Population): Model: $\frac{dN}{dt} = rN(1 - \frac{N}{K})$ N : population at time t ; K : carrying capacity (depends on resources); $r, K > 0$.

Sol: $N(t) = \frac{N_0 K e^{rt}}{K + N_0(e^{rt} - 1)} \rightarrow K$ **Equilibrium points:** $N = 0$ (unstable), $N = K$ (stable). **Quickest:** $N = \frac{K}{2}$ ($<$: 凹; $>$: 凸)

Delay Model: Model: $\frac{dN}{dt} = f(N(t), N(t-T))$ T : delay time > 0 周期解 | 与初始值无关 ps: To consider e.g. gestation period, maturation time etc.

¹Ext. of Logistic Model: $\frac{dN}{dt} = rN(t) [1 - \frac{N(t-T)}{K}]$ (用"ave. pop."估计, 准用"Convolution Type") $\Rightarrow$ ²Dimensionless: $\frac{dN^*}{dt^*} = N^*(t^*) [1 - N^*(t^* - T^*)]$ ($N^* = \frac{N}{K}$, $t^* = rt$, $T^* = rT$)

^{1,2} Periodic sol: stable limit cycle $N(t + t_p) = N(t)$, t_p : period. **Amplitude:** rT **Equilibrium Points:** $N = 0 | N^* = 0$ (unstable); $N = K | N^* = 1$ (Stable: $0 < T^* = rT < \frac{\pi}{2}$; Unstable: $T^* > \frac{\pi}{2}$)

Continuous Population Model (EN): Model: $\frac{dN}{dt} = rN(1 - \frac{N}{K}) - EN = f(N)$ $r, K, E > 0$; EN : harvesting rate/unit time; E : effort measure.

Equilibrium: $N = 0$ (unstable); $N = N_h(E) = K(1 - \frac{E}{r}) > 0$ (need $E < r$, Stable) **Yield:** $Y(E) = EN \cdot E > r$: die out; else: **Max Yield:** $Y_M = \frac{rK}{4}$ ($E = \frac{r}{2}$) **Steady State:** $N_h | Y_M = \frac{K}{2}$

Continuous Population Model (Y_0): Model: $\frac{dN}{dt} = rN(1 - \frac{N}{K}) - Y_0 = f(N; r, K, Y_0)$ Y_0 : constant yield rate/unit time.

Max Yield: $Y_M = \frac{rK}{4}$ **Equilibrium:** $0 < Y_0 < Y_M$: Two positive points (unstable(小) and stable(大)). **Sensitivity:** 对于 fluctuation 此模型和 EN 模型都不稳定, 但此模型更敏感 sensitive.

Examples of $N_{t+1} = f(N_t)$: Verhulst: $N_{t+1} = rN_t(1 - \frac{N_t}{K})$; $r, K > 0$. Better: **Ricker curve:** $N_{t+1} = N_t e^{r(1 - \frac{N_t}{K})}$

Imperfect Bifurcation: $\dot{x} = rx - x^3 + h$ (codimension-2 bifurcation) If $r \leq 0$, x^* unique. If $r > 0$, $|h| > h_c$, x^* unique; $|h| < h_c$, x^* three. (两边 stable, 中 unstable). Bifurcation points: $h = \pm h_c = \pm \frac{2r}{3\sqrt{3}}$. **cusp point:** $(r, h) = (0, 0)$

1.2 Summary - Stability and Bifurcation

	$\frac{dN}{dt} = f(N)$ No Limit cycle periodic solutions S : stable; U : unstable.				$N_{t+1} = N_t F(N_t) = f(N_t)$ Eigenvalue: $\lambda = f'(N^*)$			
Stability	Stable: $f'(N^*) < 0$		Unstable: $f'(N^*) > 0$		Stable: $ \lambda < 1$		Unstable: $ \lambda > 1$ (chaotic) Periodic: $ \lambda = 1$ $\begin{cases} + : \text{tangent bifurcation} \\ - : \text{period-doubling bifurcation} \end{cases}$	
Linearise	$N(t) = N^* + n(t), n(t) \ll 1 \Rightarrow \frac{dn}{dt} = \frac{dN}{dt} = \begin{cases} (\text{general}) = f(N^*) + f'(N^*)n(t) \\ (\text{delay}) = \text{直接带入 } t \text{ 后近似} \end{cases}$				Let $N_t = N^* + n_t, n_t \ll 1 \Rightarrow N^* + n_{t+1} = \dots = f(N^* + n_t) \approx f(N^*) + f'(N^*)n_t \Rightarrow n_{t+1} = \lambda n_t$			
Bifurcation \\\ Perturbations at N^* Periodic	Saddle-Note (Normal Form) $\dot{x} = r \pm x^2$ $0 \rightarrow S \rightarrow S-S$	Transcritical $\dot{x} = rx - x^2$ $S-U \rightarrow U-S$	(super-)Pitchfork $\dot{x} = rx - x^3$ $S \rightarrow S-U-S$	Sub-critical Pitchfork $\dot{x} = r + x^3$ $U \rightarrow U-S-U$	$\lambda < -1$: $\uparrow$ oscillations $-1 < \lambda < 0$: $\downarrow$ oscillations $0 < \lambda < 1$: $\downarrow$ Monotone $\lambda > 1$: $\uparrow$ Monotone			
	If $f^p(N^*) = N^*$, N^* is p -periodic. $\mid$ $\forall$ even p -periodic N^* with $\lambda = -1$ branches into $2p$ -periodic					Sarkovskii's Theorem: If a 3-periodic N^* exists, then $\forall p \geq 1$, p -periodic N^* exists. Species extinction: $N_{\min} < 1$		
Other	Time-Independent: At N^* : $\frac{dN}{dt} = 0$ $N(t) = N(t-T) = N^*$ $f(N^*) = 0$				Max/Min: [如果单峰 (unimodal)] 2-periodic $f'(N_m) = 0, N_{\max} = f(N_m); N_{\min} = f(N_{\max})$			
Other	Tangency Condition: If $\dot{x} = f(x, r)$, 分岔点 (x^*, r_c) : $\frac{\partial f}{\partial x}(x^*, r_c) = 0$ $f(x^*, r_c) = 0$ $\mid$ 两点合并: coalesce .							

Bifurcation: Qualitative changes in dynamics are called bifurcations. (当平衡点性质 | 个数发生变化时) **Bifurcation Diagram:** y 轴: 平衡点; x 轴: 控制参数; 虚线 unstable, 实线 stable.

稳定性 Stability 判别 | For Continuous Model: Find perturbation: $n(t)$ and $\frac{dn}{dt} \Rightarrow$ Assume $n(t) = e^{\lambda t} | ce^{\lambda t} \Rightarrow$ Find $\lambda = g(\lambda) \Rightarrow$ Let $\lambda = u + iw \rightarrow u = g_1(u)$ and $w = g_2(w) \Rightarrow u < 0$ asy.stable; $u > 0$ unstable. **Find T_c :** (如要寻找参数 (e.g. T) 范围/临界值 使得平衡点稳定) Check $T = 0$ stable. $\Rightarrow$ Using $u = 0$ (i.e. $\lambda = iw$) to find T_c .

2 Interacting Populations

Lotka-Volterra Model (Predator-Prey): $\frac{dN}{dt} = N(a - bP)$; $\frac{dP}{dt} = P(cN - d) \Rightarrow$ nondim: $\frac{dN}{dt} = u(1 - v)$; $\frac{dP}{dt} = \alpha v(u - 1)$ **Equilibrium:** $(u^*, v^*) = (0, 0), (1, 1)$

Exact sol: $\alpha u + v - \ln(u^\alpha v) = H$, $H > H_{\min} = \alpha + 1$ constant. (closed phase) **Conservation Law:** $\frac{dH}{dt} = 0$, here it satisfies. (By chain rule)

Show Level curves (constant of motion) closed: If $H(u, v)$ is level curve, Look $\begin{bmatrix} H_{uu} & H_{uv} \\ H_{vu} & H_{vv} \end{bmatrix}_{(u^*, v^*)} \Rightarrow$ If $+/-$ definite (i.e. $\forall \lambda_i > 0 (< 0)$)

$\Rightarrow$ It's local min/max at (u^*, v^*) & strictly convex/concave $\Rightarrow$ closed curves. Closed curves $\Rightarrow$ periodic sol.

Determinant Stability: For model: $\begin{bmatrix} u(t) \\ v(t) \end{bmatrix} = \begin{bmatrix} f(u, v) \\ g(u, v) \end{bmatrix}$ at equilibrium (u^*, v^*) . Let $\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} u - u^* \\ v - v^* \end{bmatrix}$, then $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} f_u & f_v \\ g_u & g_v \end{bmatrix}_{(u^*, v^*)} \begin{bmatrix} x \\ y \end{bmatrix} \Rightarrow$ Find eigenvalues λ of J by $\det(J - \lambda I) = 0$

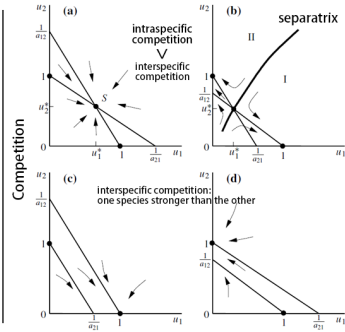
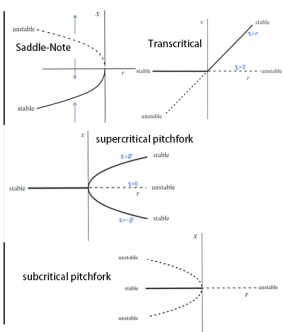
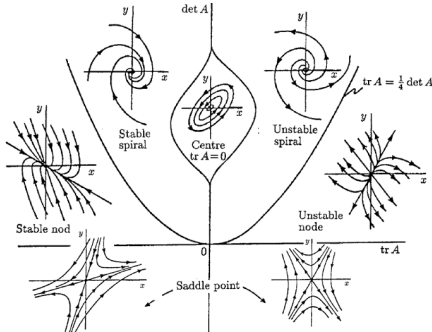
¹ $\det(J) > 0$, $\text{tr}(J) < 0 \Rightarrow \Re(\lambda) < 0 \rightarrow$ stable node $\text{Im}(\lambda) = 0$; stable focus=spiral point $\text{Im}(\lambda) \neq 0$; ² $\det(J) > 0$, $\text{tr}(J) > 0 \Rightarrow \Re(\lambda) > 0 \rightarrow$ unstable node $\text{Im}(\lambda) = 0$; unstable focus=spiral point $\text{Im}(\lambda) \neq 0$

³ $\det(J) < 0 \Rightarrow \text{Im}(\lambda) = 0$, $\lambda_1 < 0 < \lambda_2 \rightarrow$ saddle point (unstable); ⁴ $\det(J) > 0$, $\text{tr}(J) = 0 \Rightarrow \lambda = \pm iw$ (pure imaginary) $\rightarrow$ center (neutrally stable)

Null clines: $f(u, v) = 0$ or $g(u, v) = 0 \rightarrow$ 得 u, v 表达式是 nullclines. **Trapping Region:** A closed region R 可以用直线/null clines 构成 s.t. 所有边界的外法线方向 $\mathbf{n}$, 在对应边界上的点, 满足 $\mathbf{n} \cdot \begin{bmatrix} f(u, v) \\ g(u, v) \end{bmatrix} < 0$

Poincare-Bendixson Theorem: One unstable node/focus/spiral point + it's in trapping region $\Rightarrow$ A stable limit cycle exists. (Cannot be a saddle (since $\det(J)$ cannot be < 0))

Competition Model: principal of competitive exclusion: If two species compete for the same limited resource, one will eventually outcompete and exclude the other. (具体四种情况看图)



Mutualism|Symbiosis: (共生)

Case I: (图 a) coexist: competition not aggressive.

Case II: 两条射线 both species grow unboundedly.(all unstable)

Case II: 两条向外的线相交在一点 both species tend to a stable coexistence.

For small perturbations: If fixed point is Center or Improper Node ($GM=1$): Any small perturbation $\Rightarrow$ may change to spiral point. $\Rightarrow$ Not structurally stable.

Chaotic: Three dim or delay $\Rightarrow$ chaotic behaviour possible.

3 Reaction Kinetics

Enzyme Kinetics: e.g. $S + E \xrightleftharpoons[k_{-1}]{k_1} SE \xrightarrow{k_2} P + E$ K_i : rate of reaction; S : substrate; E : enzyme; SE : complex; P : product.

4 Appendix

$y' + p(x)y = q(x)$: **Integrating Factor:** $I(x) = e^{\int p(x)dx} \Rightarrow I(x)y = \int I(x)q(x)dx$ **Taylor Expansion:** $f(x, r)$ at (x_0, r_0) : $f(x, r) = f(x_0, r_0) + (x - x_0)f_x(x_0, r_0) + (r - r_0)f_r(x_0, r_0) + \frac{1}{2} \dots$

Binomial Series: $(1 + x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2!}x^2 + \dots$ **Linerisation:** $f(x + \Delta x) = f(x) + f'(x)\Delta x + \dots$ $e^{bx} = 1 + bx + \frac{(bx)^2}{2!} + \dots$ $\ln(a + x) = \ln(a) + \frac{x}{a} - \frac{x^2}{2a^2} + \frac{x^3}{3a^3} - \dots$

Hyperbolic Functions: $\sinh x = \frac{e^x - e^{-x}}{2}$ $\cosh x = \frac{e^x + e^{-x}}{2}$ $\tanh x = \frac{\sinh x}{\cosh x}$ $\sinh(x) = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots$ $\cosh(x) = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \dots$ $\tanh(x) = x - \frac{x^3}{3} + \frac{2x^5}{15} - \dots$